AzurESky "ensure quality of test"	**DATA INTEGRITY RISK MANAGEMENT AND PREVENTIVE MAINTANANCE PROCEDURE**		
	Document no: ASMF 7.11		
	Compiled by:	**Tshepo Maponya – Laboratory Supervisor**	
	Approved by:	**Nonhlanhla Msibi - Operations Manager**	

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Effective date	Rev status	Next review date
07/2025	00	07/2028

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1. Purpose

To ensure that all potential risks to data integrity within the laboratory are identified, documented, assessed, and mitigated through planned preventive maintenance and control measures, in accordance with ISO/IEC 17025 Clause 7.11 and ISO/IEC 17043 clause 7.5.2

2. Scope

This procedure applies to all computer systems, servers, external storage devices, software applications (e.g., LIMS, spreadsheets), and other information management systems used for data acquisition, processing, reporting, and storage in the laboratory.

3. Responsibilities

- **Laboratory Manager**: Ensures implementation of this procedure and allocates resources for preventive maintenance.
- **Laboratory Supervisor:** Reviews and approves the risk register, verifies the effectiveness of mitigation actions, and ensures records are maintained.
- **IT / Designated Technical Personnel**: Conducts preventive maintenance, system updates, and data integrity checks as per the schedule(Externally provided services)
- **All Staff:** Report any observed or suspected data integrity risks, malfunctions, or anomalies immediately to management.

4. Definitions

Data Integrity: The accuracy, completeness, consistency, and reliability of data throughout its lifecycle.

Preventive Maintenance: Scheduled actions intended to prevent system failures or data loss.

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Risk Register: A controlled record where all identified risks and mitigation actions are logged and monitored.

5. Procedure

QMWARELIMS12, SOP's, checklists, provide means for collecting, processing, recording, reporting, storage or retrieval of data. The functioning of the Laboratory Information Management System shall be validated for proper functioning of interface before implementation. Formulas, calculations and all other required entries/input data is able or activated to collect, process, record, store and report.
Any decisions regarding changes of the software configuration or modifications to be implemented, these shall be authorized, documented and validated before implementation

5.1 Identification of Data Integrity Risks

Risks are identified through routine operations, internal audits, incidents, or staff observations.

Examples include:

- Hardware failure (e.g., server crash, drive corruption)
- Software malfunction or version incompatibility
- Unauthorized access or accidental deletion of data
- Environmental factors (power surges, temperature, humidity)
- Human errors during data handling or entry

5.2 Risk Recording

All identified risks must be recorded in the **Data Integrity Risk Register** (ASMF 7.11 Register) using the following columns:

- Risk ID
- Description of Risk
- Likelihood (Low/Medium/High)
- Severity (Low/Medium/High)
- Risk Level (Calculated)
- Mitigation / Preventive Action
- Responsible Person
- Target Date
- Verification of Effectiveness

5.3 Risk Mitigation Through Preventive Maintenance

The IT/Technical personnel maintain/repair any damage as and when necessary

5.4 Verification and Review

The Lab Manager/Supervisor shall review the Data Integrity Risk Register and Biannually to verify:

- All identified risks are mitigated and closed.
- Backup systems function correctly.
- Preventive maintenance is conducted on schedule.

Results are discussed during Management Review Meetings.

6. Records

- Data Integrity Risk Register (DIRR)
- Backup Verification Records

Retention period: 10 years.

7. References

- ISO/IEC 17025:2017 – Clause 7.11 (Control of Data and Information Management)

8. Revision History

Rev. No.	Date	Details of Revision	Approved By
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